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New insights into hydration-induced creep behavior of shale: A comparison study of brittle black shale and clayey oil shale at micro-scale

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ABSTRACT

During hydraulic fracturing for shale gas development, water softening and creep deformation of shale rock are two common engineering problems that severely impair production performance. In order to understand the mineralogical and microstructural control on this mechanism from microscale, a brittle black shale from Lower Silurian Longmaxi Formation (named as LMX shale) and a clayey oil shale from Paleogene Youganwo Formation (YGW shale), which represent two members of shale family with different mineral constitutes and microstructures, were selected for hydration process under various durations (0.5 h–96 h) and temperatures (25 °C and 50 °C). The accompanied changes of microstructural and mechanical properties during hydration at microscale were investigated via nuclear magnetic resonance (NMR) and nanoindentation, respectively. Based on NMR T_2 spectra, peak T_2 time and water-infiltrated pore increment for both LMX and YGW shales rise sharply within the first 12 h, whereas remain invariable hereafter. And the values increase at higher temperature. For YGW shale, the maximum peak T_2 time is lower but the incremental porosity is higher than LMX shale, respectively. Such difference indicates an occurrence of gradually enlarged and connected pores inside LMX shale after hydration, while a greater number of disconnected pores have been developed in YGW shale, comparatively. For both shales, great strength deterioration and creep enhancement are monitored within the first 12 h of hydration. The modulus reduction could be as high as 26.27% and 80.16% after 96 h soaking at 50 °C for LMX and YGW shales, respectively. Creep strain rate sensitivity increases from 0.026 to 0.234 for LMX shale, and 0.011 to 0.099 for YGW shale with intensified softening degree. The softening mechanism for LMX brittle shale is dominated by secondary pores and cavities generated as a result of dissolution of carbonate cementation and grain detachment after water infiltration, while microstructural integrity of YGW oil shale is degraded by clay swelling. Due to different mineral constitutes and microstructures, creep deformation for clayey YGW shale is mainly mediated by dislocation movement of fine clay grains, whereas diffusion creep plays the key role in coarse-grained LMX shale. This comparative study with two distinct shale samples provides the mineralogy- and microstructure-based framework for interpreting water softening process and creep mechanism of shale, which benefits hydraulic fracture design in different shale formations so as to enhance shale oil/gas production.

1. Introduction

Given the continuously declining production of conventional oil and gas, shale oil and gas as new and major unconventional energy resources are playing an increasingly vital role in meeting the global energy need. In order to achieve economical hydrocarbon production from the tight shale reservoirs, large-scale drilling and hydraulic fracturing are prerequisite to generate complex fracture networks (Du et al., 2018). Water-based fracturing fluid is pressured into the target strata to create extensive fracture networks so as to promote oil and gas extraction (Wu et al., 2021; Zou et al., 2021). However, due to the physic-chemical interactions between fluid and rock during fracturing, shales are softened with enhanced creep behavior (Fiallos Torres et al., 2021), and the well production started to decline after the initial 8–12 months as a

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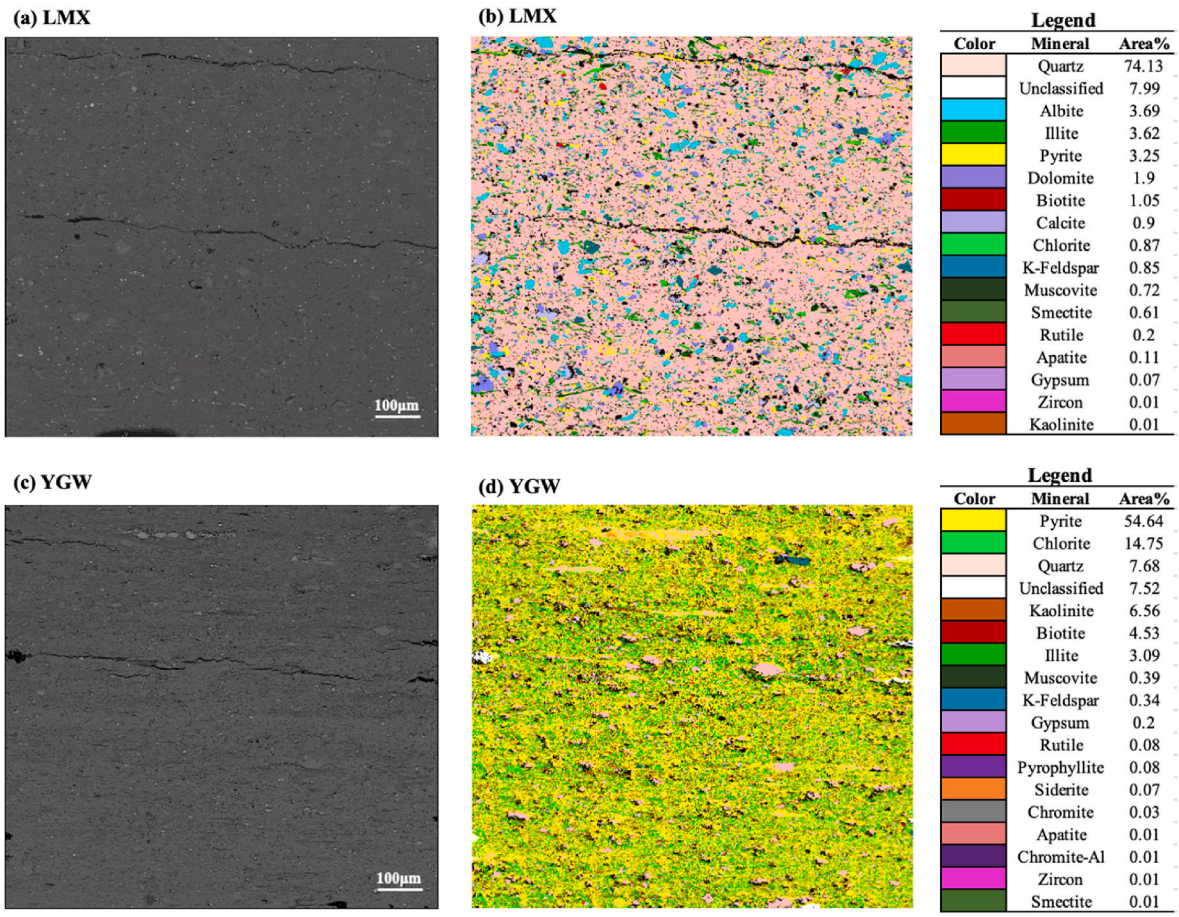


Fig. 1. Scanning electron microscopic and false-colored mineralogy images produced from QEMSCAN of LMX (a and b) and YGW (c and d). Note the pseudo-high proportion of pyrite might be a result of the extremely small grain size of pyrite but a wide distribution.

result of pertinent proppant embedment and closure of pre-induced cracks (Luo et al., 2021b). The average embedding depth of proppants into the shale can be as large as half of the proppant size (e.g., ~140 μm) under hydraulic pressure, and 88% of the fracture conductivity was lost due to water flow (Zhang et al., 2014). Although fracture closure at the early stage is governed by the elastic property of the shale, creep deformation becomes significant as time passing by. This leads to a transform of shale from an initially brittle body to a viscous material (Almasoodi et al., 2014). Such strength weakening and creep deformation can trigger multiple severe problems, not only the proppant embedment mentioned above, but also wellbore stability and reservoir subsidence, which lead to production decline and safety issues (Dou et al., 2020).

Under hydrous condition, the rheology of shale could be severely altered with changing point defect concentrations and diffusion kinetics (Mei and Kohlstedt, 2000; Wang and Jin, 2001). Strength weakening due to water softening refers to a lowered surface energy and modulus dispersion; and a reduced cohesive strength along intergranular boundaries, which takes place via detachment of silica, calcite and gypsum, as well as clay swelling (Gupta et al., 2017). Although improved understanding has been achieved on the softening process of strength decay (Fritz et al., 2020; Luo et al., 2019, 2021b; Sharma et al., 2019; Xu et al., 2020; Xue et al., 2020), the mechanisms controlling increasing creep deformation and how the mineralogical and microstructural properties affect the time-dependent plasticity after hydration treatment are still unclear. It is suggested that the physic-chemical mechanisms of shale softening are significantly dependent on the mineralogical and microstructural properties (Shi et al., 2020; Sun et al., 2020b). Shales are composed of highly compacted tiny clay particles along with interparticle nanopores and organic matter, forming a macroscopically homogeneous and continuous organoclay matrix, in which a variety of coarse-grained minerals, including quartz, feldspar, pyrite, and carbonate, are embedded as isolated solid inclusions. Comparatively, the floating brittle minerals constitute the major part in marine black shales, while clay mineral takes up more than half of the composites in lacustrine oil shales, thereby the dominated softening mechanisms are supposed to differ a lot.

Traditional macroscale bulk mechanics testing technology could not grasp such subtle and delicate softening process and reveal the mechanisms behind, especially in differentiating those among brittle and soft shale samples, as the hydration zone is generally limited within a thin exterior layer due to the extremely low permeability of shale, and the observation might be masked by the variability of shale samples. Alternatively, statistical nanoindentation measurement has been widely applied for the mechanical characterization of shales at nano-to micro scales, which is qualified in simultaneously extracting the mechanical properties of individual constituent at the nanoscale and bulk shale at microscale based on a single testing technique, while eliminating the potential errors due to samples variability (Liu et al., 2020; Luo et al., 2021a; Su et al., 2019; Sun et al., 2020a; Wu et al., 2020; Yang et al., 2020). Although such technique has been proven to be powerful in characterizing hydration-induced strength deterioration of shales (Lu et al., 2019; Luo et al., 2021b; Xu et al., 2020), little emphasis has been given to the creep behavior, and comparison on the creep deformation between brittle and soft shales after water-softening has seldomly been conducted, which hampered a better understanding of hydration-induced creep mechanism.

In this paper, a brittle marine black shale from Lower Silurian

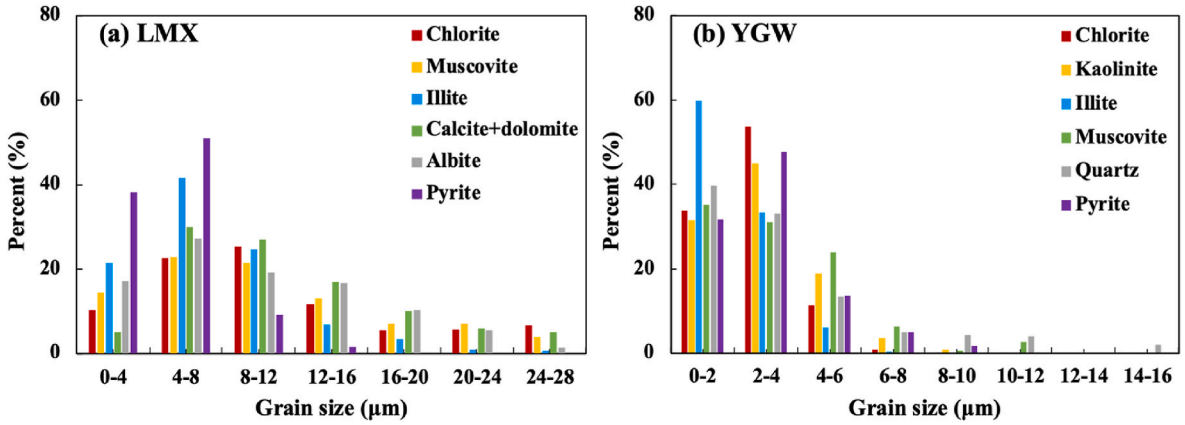


Fig. 2. Grain size distributions of different constitutive minerals for LMX (a) and YGW (b) shale samples in the areas of Fig. 1.

Longmaxi Formation in Sichuan basin, Southwest China, where shale gas has been commercially produced, and a soft lacustrine oil shale from Paleogene Youganwo Formation in Maoming basin, Southeast China with prospect for shale oil, were selected, of which the water-infiltrated pore structures, mechanical properties and creep behaviors before and after water softening were investigated via nuclear magnetic resonance (NMR) and nanoindentation, respectively. These two shales are quite different in terms of mineralogy and microstructure, thereby facilitating an comprehensive understanding of hydration-induced softening mechanisms of shale rock family. This comparative study provides the mineralogy-based framework for interpreting softening process and creep mechanism of shale, which is beneficial for hydraulic fracture design in different shale formations and shale oil/gas production enhancement.

2. Samples and methods

2.1. Samples

Two kinds of shale samples, including a brittle black shale sample from the Silurian Longmaxi Formation in Sichuan Basin (LMX shale) and an oil shale from Paleogene Youganwo Formation in Maoming Basin (YGW shale) were selected. The main mineral components for LMX shale are quartz (39.5%), albite (22.3%), illite (21.4%), calcite (7.1%), chlorite (3.9%), dolomite (3.7%), and pyrite (2.1%); and for YGW oil shale are chlorite (42.6%), quartz (21.2%), kaolinite (20.2%), smectite (10.7%), pyrite (3%), and calcite (2.3%). The mineralogical and grain size distribution within an area of 1mm × 1 mm were investigated via QEMSCAN analysis with a spatial resolution of 1 μm (Figs. 1 and 2), showing that the solid minerals of albite and carbonate are dispersed in a quartz-clay matrix in LMX shale (Fig. 1a and b), while clay minerals constitute the bedding plane of YGW shale with brittle minerals imbedded (Fig. 1c and d). Plug samples of LMX and YGW with a diameter of 2.5 cm and a length of 4 cm were prepared for NMR tests. And samples imbedded in resin with a mechanically and argon ion polished surface were prepared for nanoindentation test. Both plug samples and resin-enclosed samples were submerged in deionized water with various durations (0.5h–96h) at 25 °C and 50 °C, respectively.

2.2. Low-field NMR measurement

Low-field (LF)-NMR was performed with a Niumag MacroMR 12-150H–I analyzer. NMR T_2 spectrum were measured using Carr, Purcell, Meilboom and Gill pluse sequence, with 100 μs inter-echo spacing (TE), 10 s inter-experiment delay, 10^4 number of echoes and the minimum 200 signal to noise ratio (SNR). The initial 1st round NMR measurements were performed on original dehydrated plugs. Core plugs were then soaked in deionized water for 0.5 h at 25 °C before weighted and subjected to NMR test. This set of procedures was repeated on the same plug samples with accumulated hydration time ranging from 0.5 h to 96 h, and same for experiments conducted at 50 °C. T_2 distribution could indicate the pore size distribution for interpreting complex pore structures in shale (Ma et al., 2020; Zhou et al., 2020). Theoretically, as the bulk and diffusion relaxations are negligible for the homogeneous magnetic field with short TE, T_2 relaxation is negatively related with the ratio of surface area to pore volume, indicating a shorter T_2 relaxation

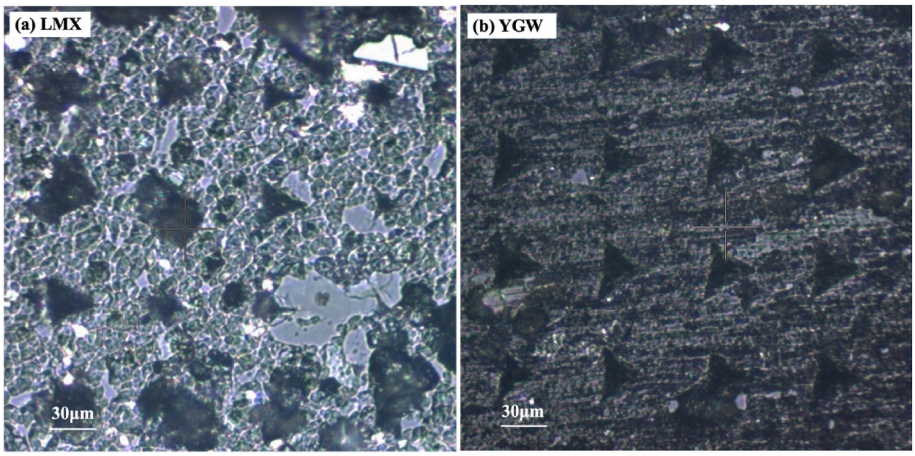


Fig. 3. Representative imprints left on original LMX shale under 300 mN (a) and YGW shale under 50 mN (b) after indentation tests.

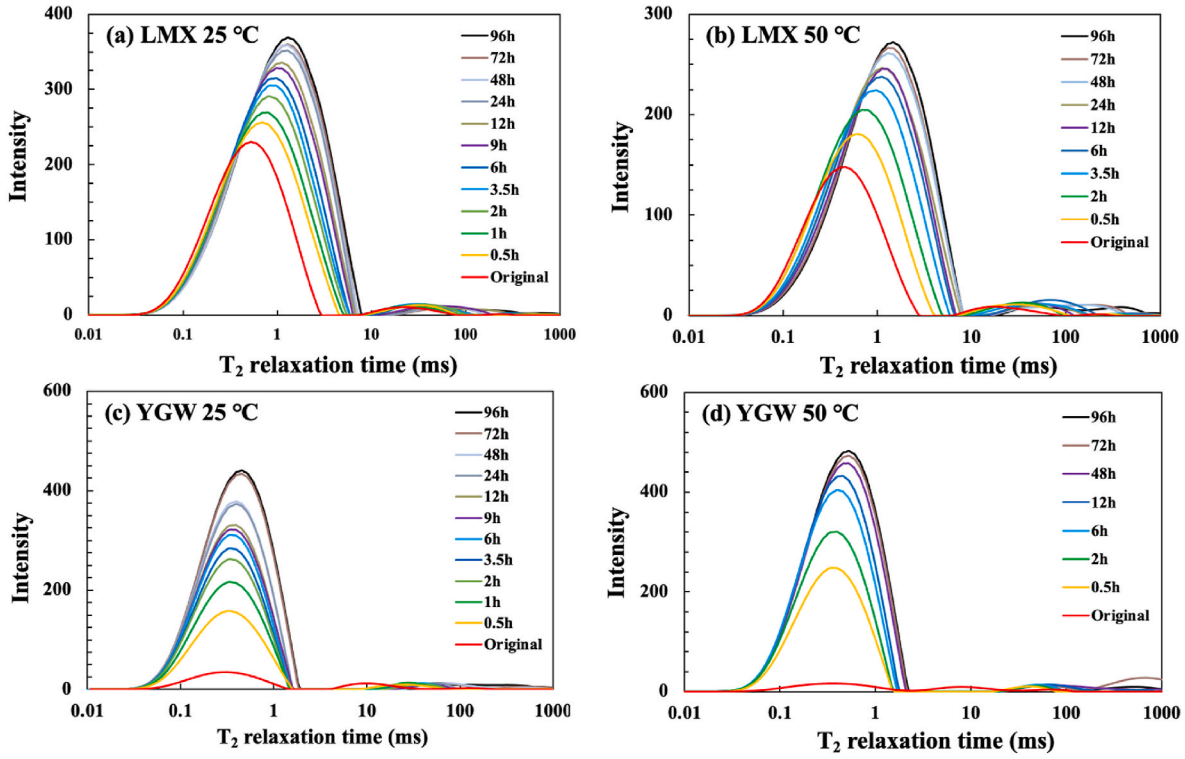


Fig. 4. NMR T_2 distributions of LMX and YGW shale samples after being hydrated at 25 °C (a and c) and 50 °C (b and d) with various time.

time corresponding to smaller pores. Here in this study, T_2 spectrum for standard samples with known porosity were measured as well for porosity estimation.

2.3. Grid nanoindentation technique

The mechanical properties, including Young's modulus and hardness, of each original and hydrated shale samples were probed by conducting 300 indents (i.e., a total of 3 matrices, each with 10 × 10 indents and a grid spacing of 70 μm) in a nanoindenter (Anton Paar TTX-NHT^3, Switzerland) with a Berkovich diamond tip. To derive the relatively homogeneous mechanical properties from the composite rather than individual phase, indentation depth (h) should be set deep enough to ensure the criterion $h/D > 0.1$ (Randall et al., 2009). As the major grain size (D) in shale samples is less than 10 μm (Fig. 2), the target maximum indentation force was set as 300 mN and 50 mN for LMX and YGW shale samples, respectively, so as to guarantee a final depth of ~4 μm, while the 30-μm wide imprints left avoided indents overlapping (Fig. 3). For each indent, the continuous stiffness measurement (CSM) method was used for loading at a constant indentation strain rate ($\dot{h}/h$) of 0.1 s^{-1} with a sinusoidal force-controlled signal of 30 mN for LMX and 5 mN for YGW in amplitude, respectively, at a frequency of 10 Hz (an example force curve of LMX is shown in Fig. S1). The maximum force was kept for 120 s during loading time, before being unloaded to 0 within 30 s linearly. Typically, CSM method outperforms in providing continuous measurement of mechanical properties along with the indentation depth during the loading period by monitoring the dynamic response of the sample-tip-indenter system caused by a small harmonic oscillation imposed on the primary monotonic loading signal (Luo et al., 2021b). Young's modulus (E) and hardness (H) for each curve can be calculated as:

$$\frac{1}{E_r} = \frac{1-\nu^2}{E} + \frac{1-\nu_i^2}{E_i} \tag{1}$$

where ν is the Poisson's ratio of sample, ν_i and E_i are elastic indentation modulus and Poisson's ratio of Berkovich indenter, respectively. E_r is the reduced modulus which can be calculated as:

$$E_r = \frac{\sqrt{\pi}S}{2\sqrt{A_c}} \tag{2}$$

where A_C is the contact area and S is the contact stiffness, which is the initial slope of the unloading branch of the force-displacement curve as $S = dF/dh$ (Fig. S1). Hardness can be obtained as:

$$H = \frac{F_{max}}{A_c} \tag{3}$$

where F_{max} is the maximum load force.

To investigate the creep behavior of shales, creep strain rate sensitivity (m), a power law exponent without units to reflect the flow ability for a material, was calculated to investigate the deformation mechanism behind (Ginder et al., 2018; Xiao and Yu, 2020). The creep strain rate sensitivity was determined by applying a power law relationship to depict the steady-state creep behavior between strain and stress (Bower et al., 1993). The brief introduction is as follows.

$$\dot{\varepsilon} = A\sigma^n \exp\left(-\frac{Q}{RT}\right) \tag{4}$$

where $\dot{\varepsilon}$ is the creep strain rate, A is a constant, σ is the applied stress, Q is the activation energy, R is the universal gas constant, T is the temperature and n is the power law (creep) exponent (which equals to $1/m$, m is the creep strain rate sensitivity). Eq. (4) could be simplified when taking T as a constant:

$$\dot{\varepsilon} = C\sigma^n \tag{5}$$

Considering the geometrical similarity of the Berkovich indenter, the strain rate and the stress can be calculated as:

$$\dot{\varepsilon} = \frac{1}{h}\frac{dh}{dt} \tag{6}$$

Table 1
Parameters based on T_2 spectrum for original and hydrated LMX and YGW shale samples.

Shale Sample	Hydration Temperature (°C)	Hydration time (h)	First Peak			Second Peak			Third Peak			Sum Area	Porosity (%)	Incremental Porosity	Damage factor
			Peak Time (ms)	Peak Area	Area Proportion (%)	Peak Time (ms)	Peak Area	Area Proportion (%)	Peak Time (ms)	Peak Area	Area Proportion (%)				
LMX	25	0	0.46	7403.34	96.77	22.22	232.05	3.03	235.43	14.77	0.19	7650.16	1.69	0.00	0.00
		0.5	0.60	8909.60	96.92	31.44	269.85	2.94	622.26	12.95	0.14	9192.40	2.16	0.28	0.17
		1	0.79	9475.93	97.19	25.53	256.21	2.63	471.38	17.59	0.18	9749.73	2.33	0.38	0.22
		2	0.79	10304.98	96.88	33.70	317.49	2.98	471.38	14.06	0.13	10636.53	2.61	0.55	0.28
		3.5	0.85	10828.78	96.69	31.44	349.02	3.12	289.94	22.11	0.20	11199.90	2.78	0.65	0.32
		4.5	0.85	11146.93	97.33	33.70	284.85	2.49	333.13	21.00	0.18	11452.78	2.86	0.69	0.33
		6	0.98	11437.66	97.62	31.44	268.10	2.29	622.26	10.63	0.09	11716.39	2.94	0.74	0.35
		9	0.98	12098.49	97.35	51.11	312.42	2.51	357.08	16.47	0.13	12427.38	3.16	0.87	0.38
		12	1.12	12372.53	97.29	44.49	321.40	2.53	439.76	22.79	0.18	12716.71	3.25	0.93	0.40
		24	1.29	13209.87	98.33	58.73	218.85	1.63	410.27	5.51	0.04	13434.23	3.47	1.06	0.43
		48	1.29	13523.00	97.67	83.10	304.83	2.20	471.38	18.34	0.13	13846.17	3.59	1.13	0.45
		72	1.29	13372.46	98.90	51.11	147.12	1.09	880.49	1.84	0.01	13521.42	3.49	1.07	0.43
		96	1.29	13685.05	97.55	44.49	316.19	2.25	622.26	27.82	0.20	14029.06	3.65	1.17	0.45
	50	0	0.52	4733.11	95.36	16.83	214.02	4.31	219.64	16.30	0.33	4963.44	0.48	0.00	0.00
		0.5	0.69	6183.09	95.25	29.33	290.14	4.47	289.94	18.35	0.28	6491.57	0.84	0.74	0.24
		2	0.79	7292.00	96.09	36.12	287.32	3.79	382.75	9.53	0.13	7588.85	1.09	1.27	0.35
		3.5	0.98	8060.31	95.97	51.11	319.24	3.80	357.08	19.53	0.23	8399.08	1.28	1.66	0.41
		6	1.12	8594.14	95.67	67.48	366.48	4.08	505.26	22.12	0.25	8982.74	1.41	1.94	0.45
		12	1.23	8759.05	97.31	51.11	238.30	2.65	252.35	3.40	0.04	9000.76	1.42	1.95	0.45
		24	1.32	9230.90	97.27	54.79	249.46	2.63	622.26	9.69	0.10	9490.04	1.53	2.18	0.48
		48	1.38	9676.69	96.23	155.22	359.22	3.57	580.52	19.87	0.20	10055.77	1.66	2.46	0.51
		72	1.48	9614.18	95.87	191.16	393.48	3.92	541.59	20.29	0.20	10027.95	1.66	2.44	0.51
		96	1.48	9784.42	96.54	51.11	200.20	1.98	382.75	150.72	1.49	10135.35	1.68	2.50	0.51
Shale Sample	Hydration Temperature (°C)	Hydration duration (h)	First Peak			Second Peak			Third Peak			Sum Area	Porosity (%)	Incremental Porosity	Damage factor
			Peak Time (ms)	Peak Area	Area Proportion (%)	Peak Time (ms)	Peak Area	Area Proportion (%)	Peak Time (ms)	Peak Area	Area Proportion (%)				
YGW	25	0	0.322	996.88	78.73	9.66	231.21	18.26	126.04	38.04	3.00	1266.12	0.06	0.00	0.00
		0.5	0.32	4568.58	95.88	25.53	184.22	3.87	943.79	11.88	0.25	4764.68	0.41	0.73	0.73
		1	0.35	6199.46	96.64	27.36	202.83	3.16	821.43	12.94	0.20	6415.23	0.57	1.07	0.80
		2	0.35	7463.35	97.82	38.72	164.17	2.15	1084.37	1.99	0.03	7629.51	0.69	1.32	0.83
		3.5	0.35	8002.13	97.87	33.70	170.22	2.08	357.08	3.97	0.05	8176.32	0.75	1.44	0.85
		6	0.35	9194.48	98.11	41.50	176.98	1.89	/	/	/	9371.46	0.87	1.69	0.86
		9	0.35	9314.07	98.50	47.69	142.03	1.50	/	/	/	9456.11	0.87	1.70	0.87
		12	0.40	9461.28	98.50	44.49	143.91	1.50	/	/	/	9605.19	0.89	1.73	0.87
		24	0.40	10905.74	97.25	67.48	274.56	2.45	541.59	34.38	0.31	11214.68	1.05	2.07	0.89
		48	0.40	12408.61	98.72	47.69	160.86	1.28	/	/	/	12569.47	1.19	2.35	0.90
		72	0.46	12868.45	97.41	58.73	268.51	2.03	471.38	73.14	0.55	13210.10	1.25	2.49	0.90
		96	0.46	13101.14	97.85	54.79	186.14	1.39	117.59	101.22	0.76	13388.49	1.28	2.55	0.91
	50	0	0.35	532.33	67.51	7.84	202.45	25.68	72.33	53.71	6.81	788.50	0.01	0.00	0.00
		0.5	0.35	7082.68	97.97	41.50	146.63	2.03	/	/	/	7229.31	0.65	1.34	0.89
		2	0.40	9129.61	98.55	47.69	134.71	1.45	/	/	/	9264.32	0.86	1.76	0.91
		6	0.40	11703.58	98.33	51.11	197.99	1.66	1335.45	1.24	0.01	11902.81	1.12	2.31	0.93
		12	0.46	12627.41	97.46	62.95	290.34	2.24	541.59	39.01	0.30	12956.76	1.22	2.53	0.94
		48	0.52	13784.17	96.86	72.33	387.81	2.73	880.49	58.82	0.41	14230.79	1.35	2.80	0.94
		72	0.52	14334.86	94.14	58.73	203.96	1.34	666.99	688.28	4.52	15227.09	1.45	3.00	0.95
		96	0.52	14716.68	96.86	58.73	251.95	1.66	505.26	225.87	/	15194.50	1.45	3.00	0.95

Porosity is calculated based on a standard sample with known porosity. Incremental porosity is calculated based on untreated original samples. Damage factor is derived via Eq. (2).

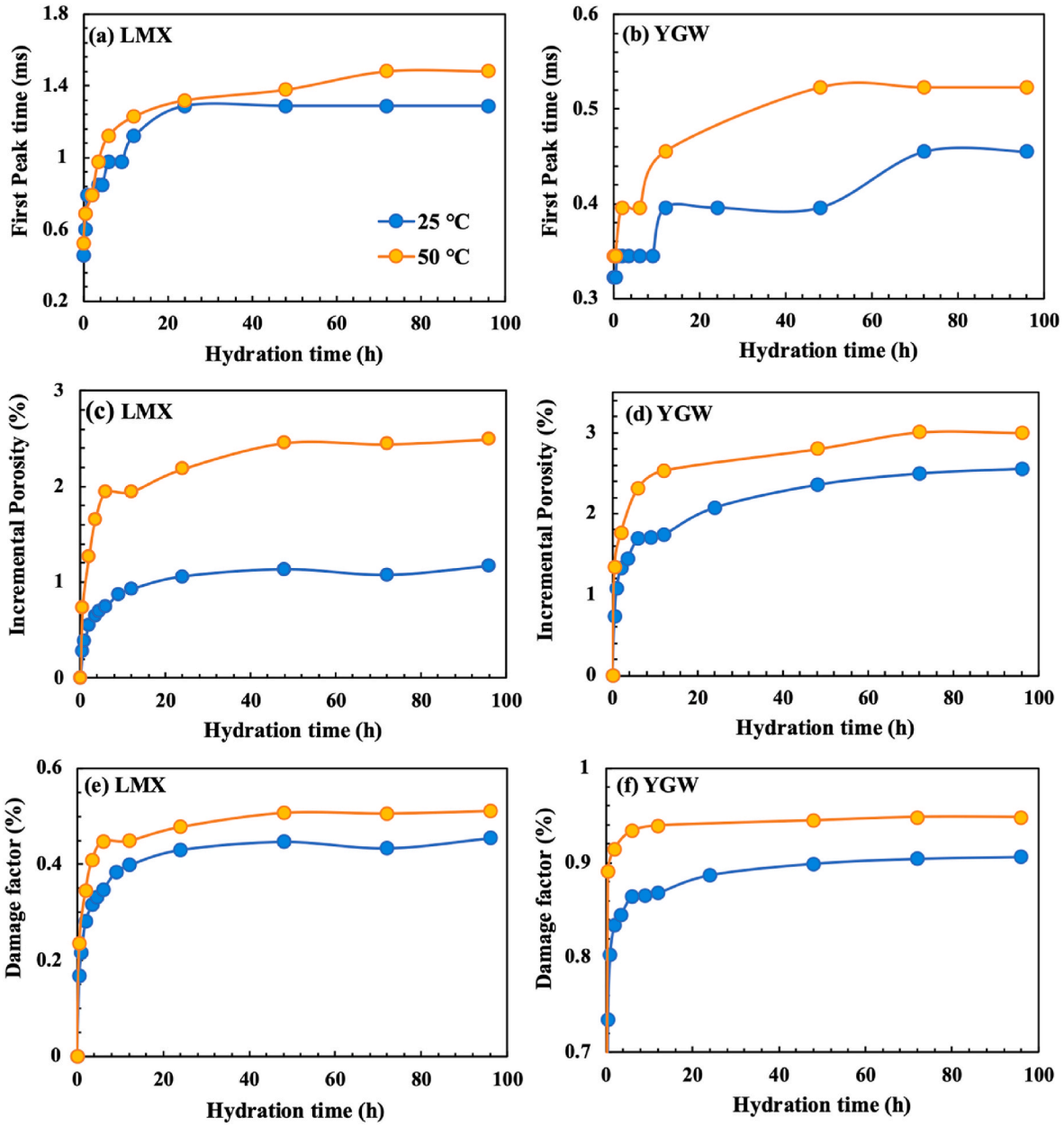


Fig. 5. Variations of the first peak time in T_2 spectra (a and b), NMR-based incremental porosity (c and d) and damage factor (e and f) with increasing hydration time for LMX and YGW shale samples.

$$\sigma = \frac{P}{24.56h^2} \tag{7}$$

where h is the indentation depth, t is the indentation time, P is indentation force. By replacing $\dot{\varepsilon}$ and σ with Eqs. (6) and (7) and taking the logarithm on both sides of Eq. (5), m can be calculated as:

$$m = \frac{1}{n} = \frac{d(log\sigma)}{d(log\dot{\varepsilon})} = \frac{d\left(log\left(\frac{P}{24.5h^2}\right)\right)}{d\left(log\left(\frac{1}{h}\frac{dh}{dt}\right)\right)} \tag{8}$$

3. Results and discussion

3.1. NMR T_2 spectrum and imaging of hydrated shales

NMR T_2 distributions of LMX and YGW shale samples after being hydrated at 25 °C and 50 °C with various durations are presented in Fig. 4, which typically display unimodal T_2 patterns (Table 1 for details). For brittle LMX shale sample, the predominant T_2 distribution ranges mainly from 0.4 ms to 10 ms, of which the peak area and time are enhanced simultaneous with increasing hydration time (Fig. 4a and b). Besides, isolated small peaks are observed above 10 ms, suggesting the presence of disconnected pores or micro-fractures in larger pore scales. Differently, longer soaking time mainly leads to a larger peak area for YGW oil shale, whereas the alternation on peak time is relatively subtle (Fig. 4c and d). The major peak time for LMX shale increases sharply during the first 12 h (Fig. 5a), while remains steady hereafter (1.12–1.48 ms), indicating gradually enlarged pores and increasingly developed cracks inside. Comparatively, the T_2 peak time for YGW oil shale rises in step within the first 12 h and then keeps at a relatively low level at 0.40–0.52 ms (Fig. 5b), implying slight pore enlargement and connection after water softening. In general, pores in brittle LMX shale are easier to be enlarged and become more connective after hydration than clayey YGW shale. By referring to the standard sample with known

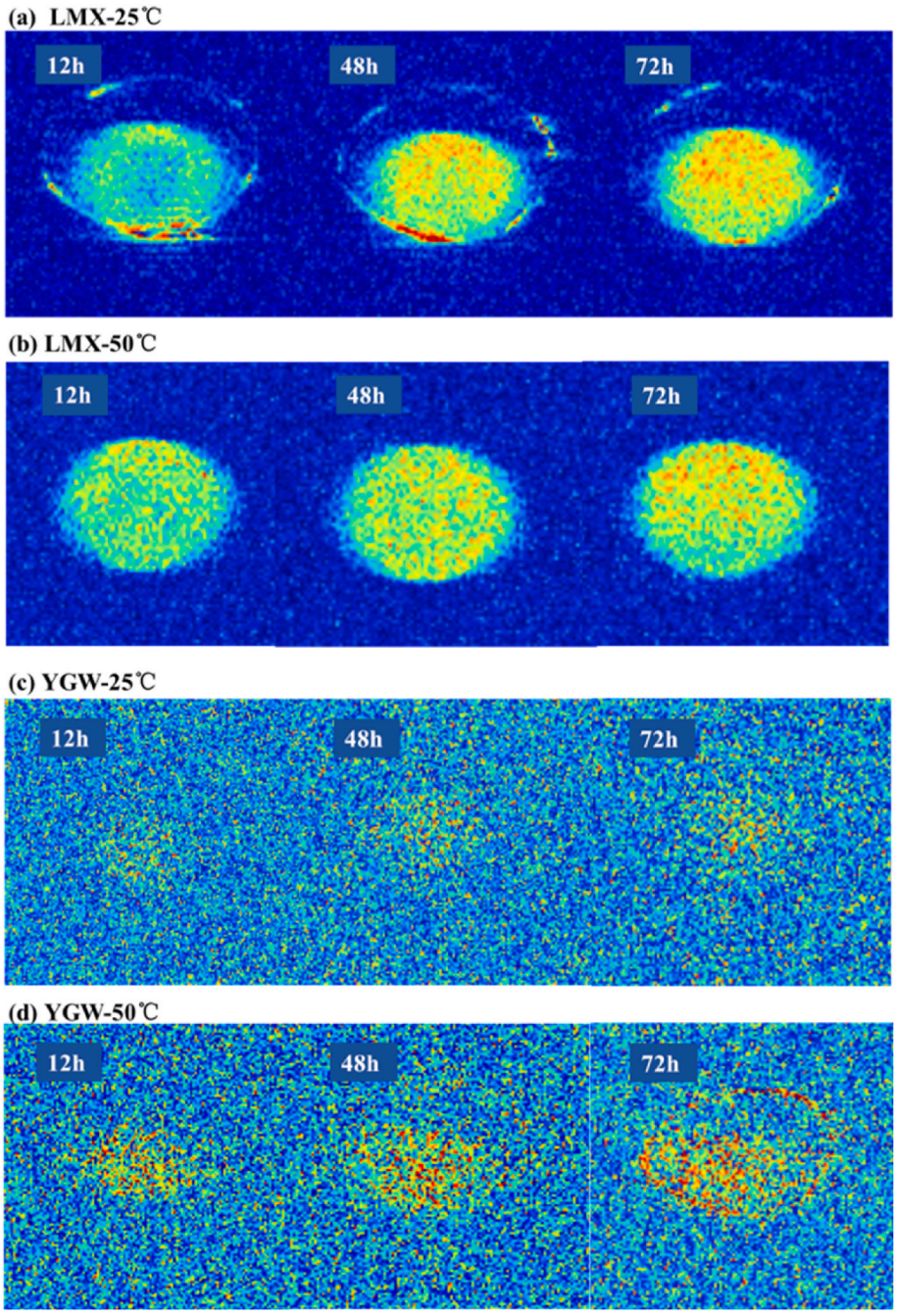


Fig. 6. NMR images of LMX and YGW shale samples after soaked at 25 °C and 50 °C with different durations.

relationship between peak area and porosity, the incremental porosities (compared to the original shale samples) were estimated (Fig. 5c and d). Similarly, both LMX and YGW show a sharp increase in the first 12 h, followed by a stable evolutionary trend. In general, the incremental porosity of YGW shale has long been higher than that of LMX shale, indicating a greater number of disconnected pores have been developed within YGW shale. For both shales, water-infiltrated pores are significantly influenced by temperature, as larger pores and increased porosity can be postulated from T_2 spectra when temperature is increased from 25 °C to 50 °C.

In order to quantify the damage caused by hydration process, a damage factor, D, referred from (Ma and Chen, 2014) was derived as:

$$D = 1 - \frac{S_{T_0}}{S_{T_2}} \tag{9}$$

where S_{T_0} and S_{T_2} are the peak area of original shale sample and after hydration, respectively. $D = 0$ is the initial state before being hydrated, while $D = 1$ represents at the limited state for being cracked. Then the variation of D with soaking time are shown in Fig. 5e and f. The values of D come to the plateau after 12 h-soaking for LMX ($D = 0.45$ at 25 °C and 0.5 at 50 °C), and 9 h-soaking for YGW ($D = 0.9$ at 25 °C and 0.95 at 50 °C), as the clayey oil shale was more vulnerable to hydration damage than brittle shale.

Representative fake-colored NMR images of plug sections are shown in Fig. 6, of which the most red and blue color represent the highest and lowest water content, respectively. Apparently, microcracks propagated under longer hydration treatment and higher heating temperature. The NMR imaging quality for LMX is far better than YGW shale, which is probably due to their different hydration mechanisms. That is, in brittle

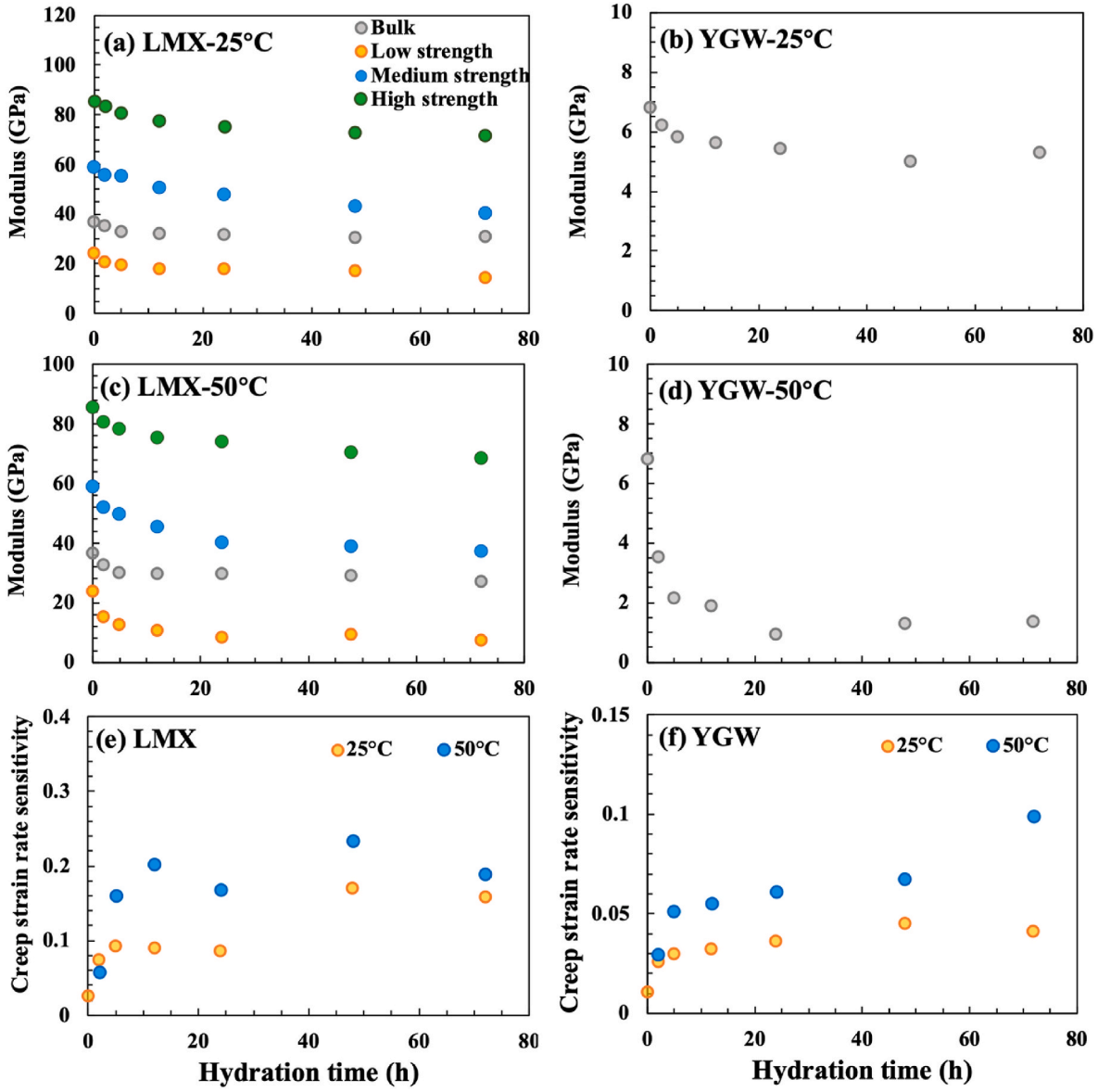


Fig. 7. Variations of modulus (a–d) and creep strain rate sensitivity (e and f) with hydration time at 25 °C and 50 °C for LMX and YGW shale samples. Phases with high, medium, and low strength were identified in brittle LMX shale, whereas YGW oil shale was analyzed as a whole.

LMX shale, the water mainly existed as free water and preserved in the fractures caused by carbonate dissolution (Luo et al., 2021b); while the osmotic swelling of clay could capture large quantities of water molecules via complex physic-chemical interactions, making it less recognizable.

3.2. Nanoindentation result

3.2.1. Data analysis

The big data analytical method proposed by (Luo et al., 2021a) was adopted to get homogenized nanoindentation results. Basic data processing procedures are schematically summarized here. For highly heterogeneous materials such as shale, the results directly collected from the grid nanoindentation tests are highly scattered and quite chaotic (Luo et al., 2021a, 2021b). Fig. S2 shows the unprocessed variations of Young's modulus and hardness with penetration depth for original LMX and YGW shale samples. The *E*-*h* and *H*-*h* curves are dispersed over a wide range at $h < 1$ μm, but gradually converge within a narrow range at $h > 1$ μm, which is a result of "Indentation surround effect". "Indentation surround effect" is profound in nanoindentation measurements on heterogeneous composites, that is the results can be severely affected by surrounding materials which are inside the elastic-plastic zone under the tip load, as indentation behavior strongly depends on penetration depth, length scale of different phases, and respective constitutive law of the constitutions.

Mechanical properties of individual constituent minerals are probed when the indentation depth is small (e.g., $h < 1$ μm). And phases with distinct mechanical properties can be identified. Then, a 3-step framework consisting of (1) data extraction and statistical deconvolution, (2) data reassembly and (3) surround effect modeling was adopted to extract the mechanical properties of individual phases as well as the large-scale bulk rock. The dataset of *E* and *H* at $h = 1$ μm in Fig. S2 are extracted and shown in Figs. S3a and b, which was statistically analyzed by Gaussian mixture modeling (GMM) (Wilson et al., 2018), and three major groups with distinct properties from high to low strength were objectively classified, including QFP (i.e., quartz, feldspar, and pyrite), carbonate and clay matrix (with minor organic matter). The clay matrix, rather than the clay mineral merely, is specifically defined as a homogenized composite consisting of highly compacted platy particles of different clay minerals with organic matter imbedded, relatively small

Table 2
Nanoindentation-based mechanical properties of original and hydrated LMX and YGW shale samples.

Hydration temperature (°C)	Hydration time (h)	LMX black shale												YGW oil shale							
		Bulk rock						Low strength		Medium strength		High strength		Bulk rock							
		E (GPa)	E Reduction (%)	H (GPa)	H Reduction (%)	Creep rate (nm/s)	m	m Enhancement (%)	E (GPa)	E Reduction (%)	E (GPa)	E Reduction (%)	E (GPa)	E Reduction (%)	E (GPa)	E Reduction (%)	H (GPa)	H Reduction (%)	Creep rate (nm/s)	m	m Enhancement (%)
Original	0	36.50		1.94		0.31	0.026		23.83		58.67		85.35		6.80		0.16		0.25	0.011	
25	2	34.86	4.50	1.59	17.82	0.45	0.054	110.89	20.47	14.10	55.30	5.74	83.20	2.52	6.20	8.82	0.15	3.21	0.57	0.026	144.34
	5	32.57	10.76	1.53	21.17	0.91	0.092	257.59	19.33	18.88	54.90	6.43	80.50	5.68	5.80	14.66	0.13	16.67	0.65	0.030	180.19
	12	31.73	13.06	1.35	30.21	2.72	0.089	247.47	17.45	26.77	50.32	14.23	77.40	9.31	5.61	17.53	0.14	13.46	0.72	0.032	205.66
	24	31.28	14.30	1.44	25.87	1.73	0.086	233.46	17.52	26.48	47.56	18.94	74.80	12.36	5.42	20.31	0.13	15.38	0.80	0.036	238.68
	48	30.12	17.48	1.40	28.03	2.36	0.171	563.42	16.89	29.12	42.74	27.15	72.64	14.89	4.98	26.79	0.13	17.31	1.05	0.045	323.58
	72	30.59	16.17	1.38	28.58	2.97	0.158	514.79	14.20	40.41	40.24	31.41	71.43	16.31	5.30	22.12	0.14	10.26	0.94	0.041	287.74
50	2	32.71	10.37	1.36	29.76	0.27	0.057	121.79	15.23	36.09	51.90	11.54	80.40	5.80	3.50	48.53	0.14	12.18	0.64	0.030	178.30
	5	30.05	17.67	1.31	32.39	1.27	0.160	521.01	12.46	47.71	49.53	15.58	78.20	8.38	2.15	68.41	0.12	23.08	1.20	0.051	383.96
	12	29.59	18.94	1.26	35.15	1.35	0.202	685.99	10.66	55.27	45.26	22.86	75.20	11.89	1.89	72.24	0.09	41.67	1.32	0.055	417.92
	24	29.48	19.22	1.28	34.19	2.85	0.167	551.36	8.42	64.67	40.12	31.62	73.80	13.53	0.91	86.59	0.10	35.26	1.48	0.061	473.58
	48	28.96	20.65	1.29	33.46	2.48	0.234	810.51	9.11	61.77	38.74	33.97	70.30	17.63	1.29	81.01	0.10	37.18	1.68	0.068	537.74
	72	26.91	26.27	1.28	34.01	3.46	0.190	637.35	7.23	69.66	37.28	36.46	68.40	19.86	1.35	80.16	0.09	40.38	2.61	0.099	833.96

E: Young's modulus; *H*: hardness; Creep rate is the minimum value of dh/dt at steady creep stage in Fig. 9; *m*: creep strain rate sensitivity. The percentage reduction in *E* and *H*, as well as enhancement in *m* is calculated based on untreated original samples.

carbonate particles as cementing agent in clay matrix, and the nanopores in clay particles.

Consequently, the new *E*-*h* and *H*-*h* curves can be generated for each constituent by reassembling the means of *E* and *H* of each Gaussian distribution along the indentation depth (Figs. S3c and d). When $h < 1$ μm, the mechanical properties of phases with strong and medium strength, including QFP and carbonate exhibit continuously declining trends with increasing indentation depth. However, clay matrix with low strength shows a similar but opposite depth-dependent behavior (i.e., increasing with depth). A unified single phase with approximately constant or depth-independent mechanical properties is achieved when $h > 1$ μm, which reflects the combined responses of all phases to represent bulk properties, as the elastic-plastic zone at this indentation depth is large enough to encompass a representative volume of bulk sample with all different individual phases included.

Then the mechanical properties of the clay matrix and its subsequently derived bulk shale are fitted by the indentation surround effect model (Luo et al., 2021b):

$$M = M_b + \frac{M_i - M_b}{1 + (h/a)^Y} \tag{10}$$

where *M* is a kind of mechanical property (Young's modulus or hardness) determined by nanoindentation at a specific indentation depth of *h*; M_i and M_b are the pertinent mechanical properties of individual phase and bulk rock, respectively. The constants *a* and *Y* are determined by curve fitting. The fitted curves of modulus along with indentation depth for major phases in original shale samples are shown in Figs. S3c and d. This model enables a smooth and progressive transition from individual phase-dominated response to bulk rock-dominated one with increasing indentation depth, namely "cross-scale characterization". Mechanical properties can be accurately determined even at extremely small depths (i.e., 0.01 μm) when the effect of surface roughness is unavoidable; conversely, the bulk rock's properties at large depth of >10 μm can be postulated, in spite of the instrumental limitation as a penetration depth of 10 μm may not always be achievable. Since the clay matrix is the major load-bearing medium in YGW shale, understanding how the mechanical properties of clay matrix change before and after hydration is the most crucial for assessing the degree of softening; and given the relatively low and narrow range of strength for YGW, it is analyzed as a whole without further phase differentiation (Fig. S4), while three phases are still identified for LMX brittle shale.

3.2.2. *Influence of hydration on mechanical properties*

Based on Fig. 7a–d and Table 2, the mechanical properties of both bulk and individual phases significantly decreased upon increasing hydration time, and higher temperature could accelerate the softening process intensively. Modulus reduction of LMX bulk rock could be as high as 26.27% (50 °C, 72 h), and the softening degree for individual phase decreased in the sequence of low (69.66%), medium (36.46%), and strong strength (19.86%), which is similar to previous study as the reduction of the average Young's modulus for brittle shale from Longmaxi Formation could reach up to approximately 39%, after being immersed in de-ionized water for two days (Du et al., 2018). As for the clayey YGW oil shale, although clay minerals in YGW shale sample are mainly non-swelling chlorite and illite, water softening could still reduce 80.16% of the modulus at 50 °C, 72 h, revealing that the structural integrity of clay matrix is significantly undermined by the shale-fluid interactions. This softening behavior of clay composites was also observed by previous study (Wang et al., 2021). Even stiff inclusions shale, such as quartz, feldspar and carbonate are thought to be resistant to fluid interaction, they are all weakened to some degree due to hydration treatment. Because of the indentation surround effect, measurements are not only controlled by the stiffness of the indented material itself, but also affected by the stiffness contract between hard inclusions and soft surrounding clay matrix. Then the softening of the

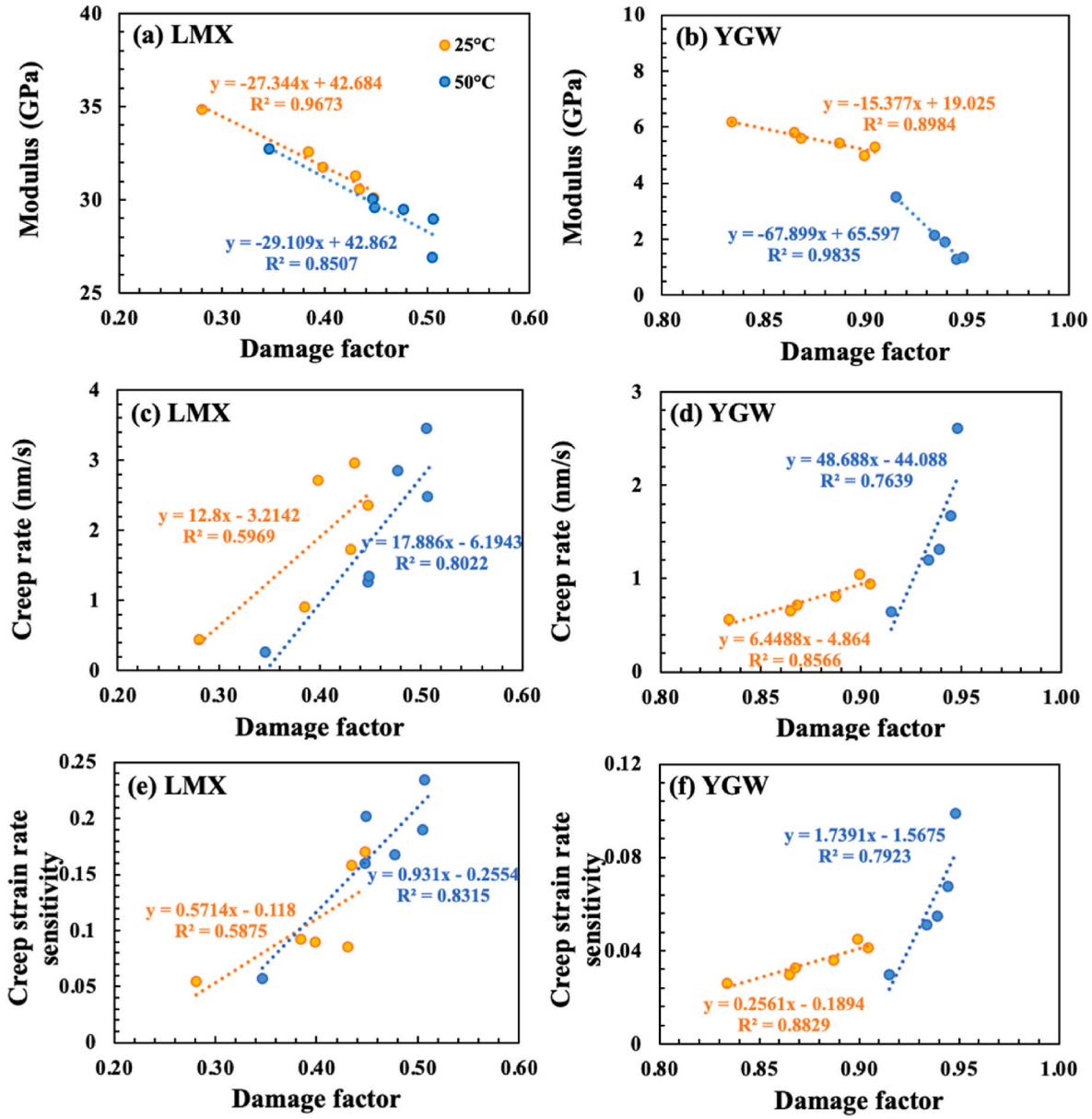


Fig. 8. Variations of modulus (a and b), creep rate at steady stage (dh/dt, t = 120s in Fig. 9) (c and d) and creep strain rate sensitivity (e and f) with damage factor for LMX and YGW shale samples.

clay matrix exacerbates the mechanical mismatch, which causes the measured mechanical properties of inclusions to be affected by the softened clay matrix literally at the beginning of the indentation loading (Luo et al., 2021a). Notably, the modulus has a negative linear correlation with NMR-based hydration damage factor (Fig. 8a and b), implying a strong control of water softening process on shale strength. The slopes for LMX at 25 °C and 50 °C are consistent with each other (Fig. 8a), while for YGW the slope at 50 °C is much higher than that at 25 °C (Fig. 8b), indicating a more temperature-sensitive mechanical property for clayey oil shale compared with brittle black shale.

3.2.3. *Influence of hydration on creep behavior*

Averaged creep displacement-time curves (without phase-differentiation) during holding period for original and hydrated LMX and YGW shale samples are shown in Fig. 9. The starting points for all tests are aligned, that is the (0, 0) point is the time that the holding period began, while the end point is the time that the holding period ended. Apparently, creep displacement increases with longer softening time and higher temperature, and YGW oil shale sample is prone to creep with much deeper creep displacement compared to LMX brittle shale. The logarithmic plots of strain rate ($\dot{\varepsilon}$) and stress (σ) for original and hydrated LMX and YGW shale samples in determining the values of m by using Eqs. (4)–(8) are shown in Fig. S5. Variations of m with hydration time are shown in Fig. 7e and f (Table 2), which display sharp increasing trends within the first ~12 h and remain steady hereafter. Brittle LMX generally possessed much higher m (0.026–0.234) than soft YGW oil shale (0.011–0.099), implying different creep mechanism occurred. According to the variations of creep rate at steady stage (dh/dt, t = 120s in Fig. 9) and m along with damage factor (Fig. 8c–f), hydration process is quite influential in affecting creep deformation of shales, as revealed by the linear positive correlations observed for both LMX and YGW shale samples. The slopes of creep rate and m for LMX vary a little between 25 °C and 50 °C, whereas increase a lot from 25 °C to 50 °C for YGW samples, implying a more temperature-sensitive creep behavior for YGW shale sample.

In summary, the big data nanoindentation technique is accurate to

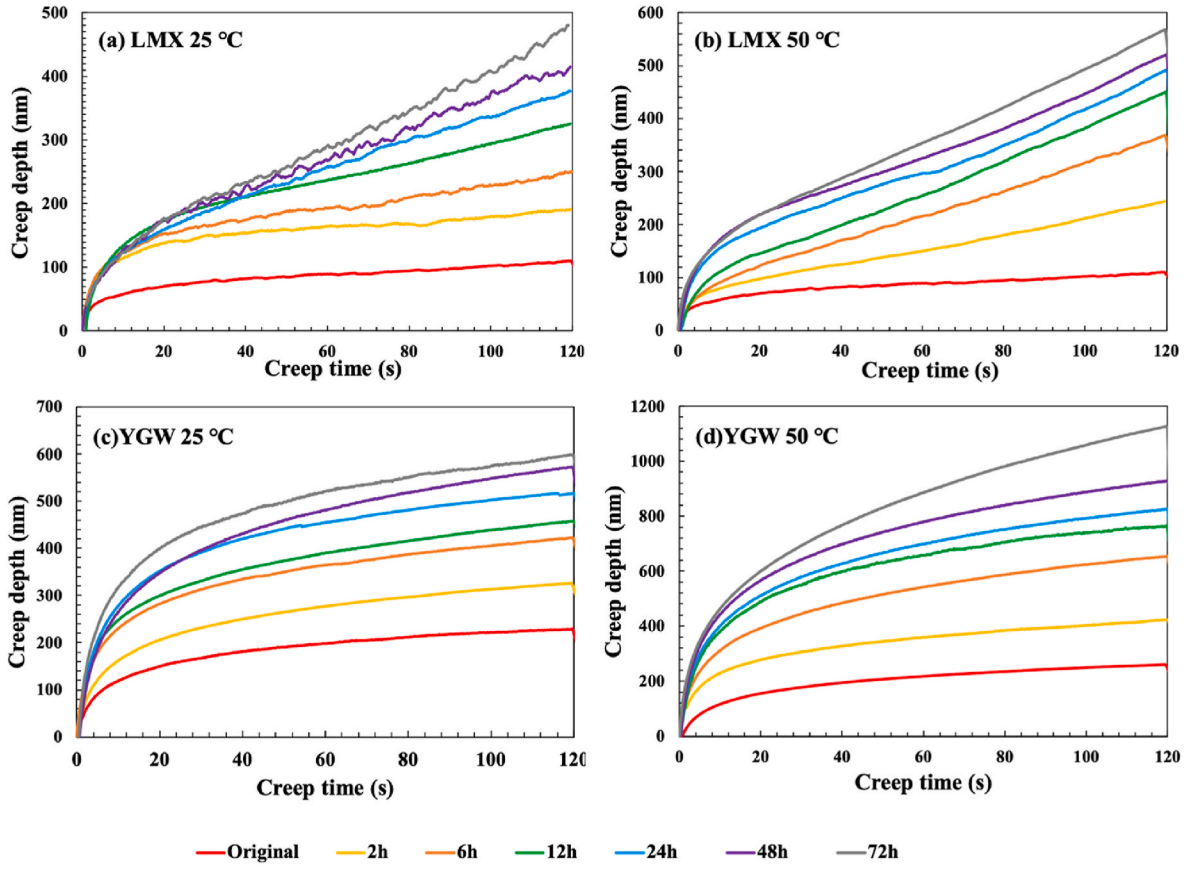


Fig. 9. Averaged creep depth v. s. Creep time for LMX and YGW shale samples at 25 °C and 50 °C with no phase-differentiation.

detect subtle changes in the mechanical properties of shale, and thus can be used as a practical technical toolkit to evaluate the degree and understand the mechanism of softening.

3.3. Softening mechanism

Considering the physic-chemical interactions during shale hydration, clay-fluid interactions, double layer repulsion and minerals dissolution are considered as three major mechanisms for shale softening (Du et al., 2018; Ma et al., 2016), and their relative contributions mainly depend on mineralogical and microstructural properties of shale samples. Based on mineralogical and microstructural features, distribution of solid minerals in clay matrix can be classified into two types, lenses network and scatter patches (Du et al., 2020). Lenses network means the solid minerals are spatially distributed in a continuous network, which works for YGW shale with aggregated solid grains floating in clay matrix (Fig. 1c and d). Comparatively, scatter patches with solid minerals distributed in an isolated pattern in clay matrix (Greg, 2002), best depict LMX shale, of which the solid grains are evenly and sporadically distributed (Fig. 1a and b). In this view, the dominating softening mechanism for two shale samples are discussed separately with respect to their mineralogical and microstructural features.

3.3.1. LMX black shale

As for LMX black shale with a high proportion of brittle minerals, large particles (mainly quartz and albite with a size of 4–12 μm, Fig. 2a) are floating in clay matrix, and the dispersed calcite behaves as the cementation agent contributing to the interparticle bonding within the clay matrix. The softening mechanism for LMX black shale is closely related to pores and cavities developed due to dissolution of carbonate cementation and grain detachment after water infiltration (Manjunath and Jha, 2019).

3.3.1.1. Pyrite oxidation and carbonate dissolution. Brittle shale usually contains cementing agents such as feldspar and carbonate minerals, which lithifies the deposition and binds the components together. A high fraction of calcite could promote interparticle cementation and hence strengthens the mechanical properties of shale, while the decreasing of the integrity of rock structure by dissolving cementing agents can significantly hamper the strength of shale. Given the ubiquitous pyrite and carbonate in LMX black shale (Fig. 1b), pyrite oxidation and the induced carbonate dissolution are supposed to be rather active during the rock-fluid interactions. Pyrite (FeS_2) would be oxidized upon exposure to oxygen and water, while producing ferric iron hydroxide [$Fe(OH)_3$] and sulfuric acid (H_2SO_4) (Golez and Kyuma, 1997). And the overall reaction is as: $4FeS_2+15O_2+14\ H_2O = 4Fe(OH)_3+8H_2SO_4$. The sulfuric acid produced in turn can dissolve carbonate cementation in shale. High concentrations of calcium, magnesium, and sulfate were detected in the supernatant after soaking a crushed shale sample in water for three weeks (Ali and Hascakir, 2015). And a 48 h-immersion of fresh cut shale from Longmaxi formation in de-ionized water demonstrated a substantial dissolution of $CaCO_3$ (Du et al., 2018). Thus, the reactions of pyrite oxidation and the following carbonate dissolution may significantly decline the strength of brittle LMX shale sample.

3.3.1.2. Grain detachment. As a consequence of carbonate dissolution, particles with a shallow embedment may lose their physical constraints at the boundaries, and were removed from sample surface by interparticle repulsive forces (Levenson and Emmanuel, 2017), which could weaken rock strength severely. It is suggested that the detachment of grains from the rock surface could accounts for as high as ~40% of the mass removal during weathering on fine-grained limestone, due to the chemical dissolution along grain boundaries (Levenson and Emmanuel, 2016). The intergranular repulsive forces are likely to be related to water adsorption on the crystal surfaces or the electric double layer, and

a reduced concentration of dissolved ion is expected to increase the extent of repulsion (Røyne et al., 2015). This kind of particle detachment during hydration treatment contributed to the formation of micrometer-sized cavities on brittle shale, and facilitated the subsequent microchannel opening for water infiltration (Luo et al., 2021b), which can be confirmed by the simultaneously enhanced T_2 peak height and time with increasing hydration time (Fig. 4a and b), revealing the gradually enlarged pores and increasingly developed cracks inside. Moreover, the similar variation trends of modulus with damage factor for LMX shale at different temperatures (Fig. 8a) also indicate the consistent dominating softening mechanism as water infiltration and physical erosion.

3.3.2. YGW oil shale

Clay matrix, mainly non-swelling chlorite and kaolinite is the major load-bearing media for tested YGW oil shale, and the relatively small sized particles (mainly quartz with a diameter less than 4 μm, Fig. 2b) are randomly embedded as isolated solid inclusions in the homogenized clay matrix. Different from brittle LMX black shale, of which the initial dissolution of cemented carbonate agent is probably the main contributor to shale softening, clay swelling induced by complex physic-chemical interactions in fluid may come to first in the instability of YGW oil shale. This can be confirmed with fact that although higher NMR T_2 peak is observed with stronger hydration, the alternation on T_2 peak time is relatively subtle (Fig. 4c and d), which implies that water infiltrated in YGW is mainly absorbed onto clay with little pore enlargement. Moreover, the declining trend of modulus with damage factor at 50 °C is much speedier than that at 25 °C (Fig. 8b), consolidating a temperature-dependent clay swelling behavior for YGW shale, as thermal motion and diffusion coefficient of interlayer spacing water increase at higher temperature (Akinwunmi et al., 2019).

Although expansive smectite takes up only 10.7% of the total in YGW shale, the large amount of chlorite (42.6%) and kaolinite (20.2%) identified in the oil shale here without interlayer swelling properties, still witnessed intensive osmotic swelling, which happened between the electrical double layers due to the ion concentration gap between the pore fluid near clay surface and in bulk pore fluid (Dominijanni and Manassero, 2012; Fritz et al., 2020; Luo et al., 2019; Sebastián et al., 2008; Xue et al., 2020). Due to the negative charges at the surface of clay particles, a distribution of cations and anions can be formed in the electrolyte between clay particles, which will give rise to the double-layer repulsive potential (Darley, 1969). The thickness of the electrical double layer associated with clay minerals is supposed to be of the same order or even thicker than the pores in shales. As far as chlorite, kaolinite and smectite concerned, the overlapping of electrical double layers associated with the charged surfaces of the clay particles, which are exposed in opposing walls of micro- or meso-pores of the shale, can bring about electrostatic repulsion and then lead to increased pore or hydration pressure (Wilson and Wilson, 2018), while connecting the pore network under water infiltration. Mechanical properties can be severely degraded by clay swelling. The water absorbed by the clay particles acts as lubricant and hence reduces the interparticle frictional resistance. Young's modulus was found to decrease with increasing swelling and the shale behaved like a ductile material gradually (Du et al., 2018).

3.4. Creep mechanism

Water is supposed to influence creep rate primarily through its effect on the concentrations of intrinsic point defects, ionic diffusion and dislocation climb (Mei and Kohlstedt, 2000). Creep mechanism mainly includes dislocation (edge and screw dislocation, kink-pair nucleation) and diffusion activities (grain boundary diffusion, grain boundary sliding and lattice diffusion), of which the creep strain rate sensitivities varied a lot (Mohammed et al., 2021; Sun et al., 2019). Total strain rate of the sample is the sum of the those accommodated by dislocation creep and diffusion creep, and these two processes are independent. The strain rate sensitivity is an important parameter in the description of the kinetic and thermodynamic characteristics that had been widely used to uncover the plastic deformation mechanisms of different metals and alloys (Herbert et al., 2018; Li et al., 2018; Mohammed et al., 2021; Sun et al., 2019; Zhao et al., 2018), which functions as identity for numerous creep mechanisms. Diffusion mechanism is generally featured by $m > 0.25$, while the dislocation one is characterized by $m < 0.25$ (Sun et al., 2019; Zhao et al., 2018). However, due to the complicated interactions between grain boundary diffusion and dislocation activities, the value of m could only be viewed as a reference in identifying the dominating creep mechanism. As shown in Fig. 10, creep behavior of YGW shale



Fig. 10. Distribution of creep strain rate sensitivities and the corresponding mechanisms for LMX and YGW shale samples.

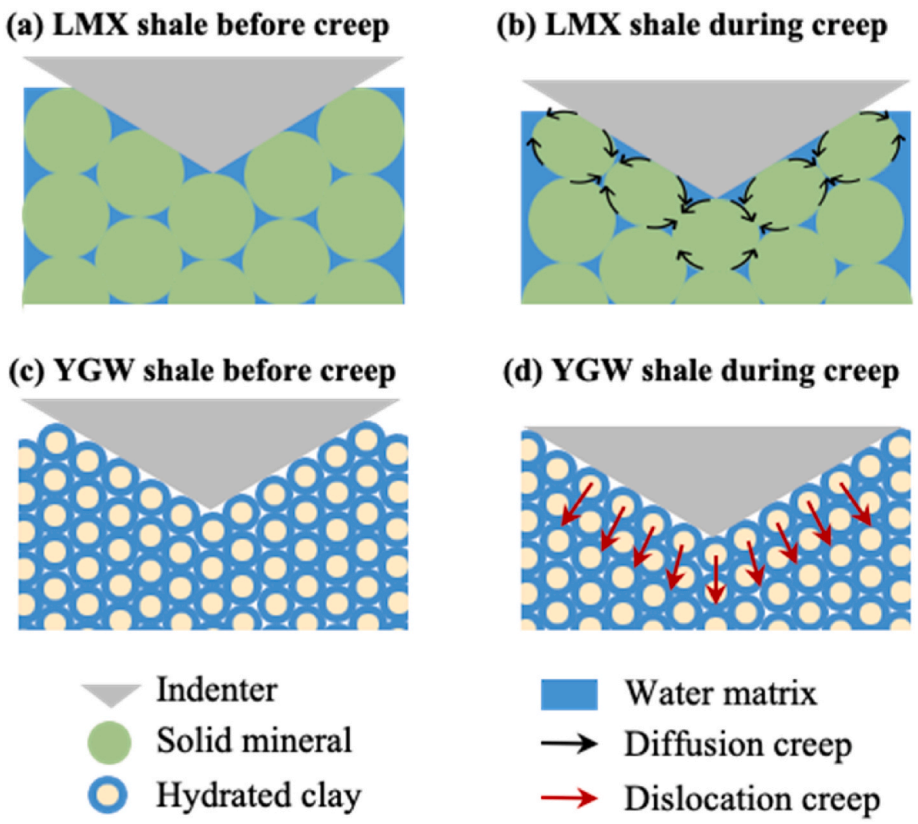


Fig. 11. Schematic diagram of creep process under indenter for LMX shale (a and b) and YGW shale (c and d). LMX shale is simplified as accumulation of solid mineral grains with voids filled up with water, and the creep is mainly due to grain boundary diffusion. YGW shale is illustrated as a packing of hydrated clay particles, of which the dislocation movement accounts for major creep deformation.

may be mainly attributed to dislocation creep, while for LMX shale sample, the increasing value of m with enhanced hydration effect indicates a potential transition from dislocation to grain boundary activity creep mechanism.

Mineralogical and microstructure characteristics play the dominating role in determining the petrophysical and rheological properties of shale. As discussed in Section 3.3, the lenses network distribution of morphology for YGW oil shale indicates that the released energy under indentation is totally dissipated in creating new surfaces of the crack, and the rheology may approach a homogeneous strain condition due to stronger phase functioning as the stress-supporting network. While for the scatter patches distribution of LMX, a plastic zone is developed under the crack tip, and the released energy is dissipated both in the plastic deformation of the plastic zone and in creating new surfaces (Guo et al., 2018), and the rheology may be more likely to a homogeneous stress condition (Greg, 2002). For both LMX and YGW shale samples, plastic deformation is mediated by nucleation and movement of intragranular dislocations, as intersecting slip planes create permanent dislocation network. During the creep process of the indentation test, excess dislocation would align themselves forming dislocation walls and consequently develop into incipient kink banks. These kink bands contributed to the creep displacement when geometrically necessary dislocation density exceeded the maximum density that can be stored, as the repulsive force acting on the contacted/compressed area forced the material to spread out (Du et al., 2020). When softening degree increases, grain boundary dislocation sources would be activated, and dislocations begin to nucleate and propagate across the grains with their two ends depinning and shifting along the grain boundaries. However, such dislocation activity is less significant for LMX shale sample with increasing hydration, while grain boundary diffusion gradually takes over (Fig. 11). Water along triple junction channels effectively shortens the grain-boundary diffusion path and provides rapid diffusion pathways (Dell'Angelo et al., 1987). Indentation induced fracture would initiate at the grain boundaries. LMX shale composed of large and connected solid grains is more likely to fracture during loading period, and the relatively clean and smooth grain boundaries may help the fracture to propagate; then at holding stage, grain boundary-mediated bulk diffusion comes into play and is supposed to take place concurrently with the dislocation activity. All these reactions, especially the diffusion creep, may be severely accelerated under intensified hydration treatment, including higher temperature and longer time.

3.5. Limitations and implications

Although most geologically applicable creep laws have been applied for rock samples, shale is not monomineralic but highly heterogeneous with grains in various morphologies and sizes. The averaged grain size for LMX (4–12 μm) is larger than that for YGW (0–4 μm), and the same trend has also been confirmed on the value of m (m = 0.026–0.234 for LMX, and 0.011–0.099 for YGW), which indicates a grain size-dependent creep behavior. The creep law applied here (Eq. (4)) ignored grain size effect, which, however, could severely influence creep behavior for clastic sediments as: $\dot{\varepsilon} = A'\sigma^n d^{-p}\exp\left(-\frac{Q}{RT}\right)$, where A' is the constant, d and p are grain size and grain size sensitivity, respectively (Greg, 2002). The other parameters are the same with Eq. (4). Rheological data on fine-grained aggregates synthesized from natural minerals with known grain sizes has allowed the analysis of creep with controlled microstructure, especially for diffusion creep in peridotite from mantle, which is highly sensitive to olivine grain size (Greg, 2002; Miyazaki et al., 2013; Wang et al., 2005). However, little work has been done on clastic rocks.

Furthermore, the results here confirmed that the mineralogy of shale

dictates the degree of shale softening. A well investigated mineralogical composition of shale is prerequisite in identifying shale softening. So far, shale has long been simplified as a two-phase (solid minerals and clay matrix) composite, and the deformation flow under indentation is dominated by the strength contrast of the two phases, the proportion and topology of the strong phase. However, more experimental and theoretical works are required on quantifying "mixing laws" with which the flow laws of the individual minerals can be used to study the properties of multi-phase mixtures.

Shale gas reservoirs are located a few thousand meters underneath where temperature, pressure and water fugacity are much higher than the atmospheric conditions, which significantly impacts the shale-fluid interactions and further controls the process of shale softening behavior, such as diffusion kinetics under hydrous condition. Therefore, concurrent with the nanoindentation study under various experimental conditions (hydration temperature, pressure, and duration), a complete mineralogical analysis should be performed to develop a predictive capability of shale softening with known mineralogy and morphology. Such framework for interpreting the softening mechanisms of shale facilitates the design of shale gas exploration and stabilization of gas production rate.

4. Conclusions

This paper provides new insights into hydration-induced creep behavior of shale by comprehensively compared the water softening processes and nanoindentation creep behaviors between two shale samples with distinct microstructures, which is of great significance in understanding rock rheology at core level in the process of hydraulic fracturing.

For brittle LMX shale, microstructures are altered with pores being gradually enlarged and connected after water infiltration; while for clayey YGW shale, increasing number of pores due to clay-swelling accounts for its main microstructural changes. Comparatively, the degree of strength deterioration and creep deformation for YGW shale is much higher than that for LMX shale as a result of mineralogical and microstructural differences. Water softening mechanism for LMX shale is dominated by pores and cavities generated by dissolution of carbonate cement and grain detachment after water infiltration, while YGW shale is mainly degraded by clay swelling. Creep deformation for clayey oil shale is mediated by dislocation movements of fine clay grains, while diffusion creep plays the key role in coarse-grained brittle LMX shale.

It is worth of mentioning that the creep law applied here ignored the highly heterogeneous nature of shale with grains in various sizes, so future study considering a grain size-dependent creep model is urgent. Besides, more work is required on quantifying "mixing laws" with which the flow laws of the individual minerals in shale can be used to study the properties of multi-phase mixtures, rather than a simplified two-phased composite. In general, this study shed light upon hydration-induced creep mechanism of shale at microscale, which is essential in preventing potential production decline during hydraulic fracturing.

Declaration of competing interest

The authors declare that they do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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Appendix A. Supplementary data

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